

Andrew J. Graven

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SUMMARY

Recent Caltech Mathematics PhD Graduate (June 2026) with experience in scientific computing, numerical optimization, and numerical astrodynamics. Built **GPU-accelerated** tools at NASA JPL for large-scale orbital analysis and mission design, including production software achieving $10^6 \times$ **speedups**.

EDUCATION

Caltech	Mathematics PhD (June 2026) Advisor: Nikolai G. Makarov Awards, Fellowships & Grants 2026 Caltech S. R. Johnson Graduate Dissertation Award (\$1,000) 2025 Caltech S. R. Johnson Prize for Excellence in Graduate Study (\$1,000) 2023 DOD NDSEG Exemplary Mathematics Poster Presentation 2021-2024 DOD NDSEG Graduate Research Fellowship (\$42,200/yr)
Cornell University	Mathematics BA (2021) <i>Summa Cum Laude</i> , Distinction in all subjects Awards, Fellowships & Grants 2018-2020 Caltech Summer Undergraduate Research Fellowship (\$7,000/summer)

WORK EXPERIENCE

Caltech – Mathematics PhD	2021 - 2026
Research in inverse problems , dynamical systems , and nonlinear PDEs , using tools from complex analysis, potential theory, and conformal mapping.	
NASA JPL – SURF Fellow	Summer 2018, 2019, 2020
2020	Delivered a production software tool (CUDA/C) for NASA to predict long-term satellite-ground station availability for satellite constellations. Achieved a $10^6 \times$ speed-up over classical algorithms. Presented results at the <i>AAS Space Flight Mechanics Conference (2021)</i> .
2019	Built and implemented a GPU-accelerated (CUDA/C) tool to scale interplanetary spacecraft tour design, accelerating evaluation/search of candidate tours in support of the <i>Europa Clipper</i> mission.
2018	Developed an ergodic theory-based approach to estimate long-term statistical behavior of satellite orbits, enabling rapid forecasting without brute-force long-time simulation . Presented results at the <i>Astrodynamics Specialist Conference (2019)</i> .

SKILLS & LEADERSHIP

Languages	Python, C, CUDA/C, MATLAB, JavaScript, Mathematica.
Systems & Tools	Linux, Windows, Git, OpenMP (Cornell CS), PyTorch (self-study), LaTeX
Expertise	Optimization, parallel computing, numerical linear algebra & machine learning (Cornell CS); scientific computing & numerical astrodynamics (NASA JPL).
Leadership	Boy Scouts: Eagle Scout Water Polo: Caltech goalie & coach; team character award established in my name.

SELECTED PUBLICATIONS

- Graven, Andrew (Apr. 2026). *Analysis of Log-Weighted Quadrature Domains*. arXiv: [2604.10394](#) [\[math.CV\]](#). URL: <https://arxiv.org/abs/2604.10394>.
- Graven, Andrew and Nikolai G. Makarov (Sept. 2025). *Quadrature Domains and the Faber Transform*. arXiv: [2509.03777](#) [\[math.CV\]](#). URL: <https://arxiv.org/abs/2509.03777>.
- Andrew Graven, Alan Barr, and Martin Lo (Jan. 2021). “A Rapid Method for Orbital Coverage Statistics With J2 Using Ergodic Theory”. In: *AAS Space Flight Mechanics Meeting*. URL: <https://arxiv.org/abs/2102.03459>.
- Leo Huang, Andrew Graven, and David Bindel (Apr. 2021). “Density of States Graph Kernels”. In: *SIAM International Conference on Data Mining*. URL: <https://arxiv.org/abs/2010.11341>.
- Andrew Graven and Martin Lo (Aug. 2019). “The Long-Term Forecast of Station View Periods for Elliptical Orbits”. In: *Astrodynamics Specialist Conference*. URL: <https://arxiv.org/pdf/2010.06021>.